

Budgeted Counterfactual Landscape Distillation for PPO Theory, Method, and Multi-Task Experimental Plan

Working research plan

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Abstract

Adding an action-dependent term to PPO’s advantage changes its actor update, whereas a state-only term has zero direct population score-gradient effect. That fact alone is not a contribution. This project asks whether a small, explicitly measured budget of expensive all-action counterfactual (CF) simulator branches can be amortized into a learned state-conditioned action landscape that guides ordinary PPO at states that were never queried. The current CF target is a one-step critic-bootstrap teacher, not an exact long-horizon oracle. We train an ensemble of landscape students, use its mean as a confidence-gated action-selective advantage perturbation, and use its prediction variance to choose later CF queries. The plan validates the mechanism, the teacher, sparse-label amortization, and active allocation across distinct sparse-reward MiniGrid task structures.

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1 Question, scope, and claim boundaries

The research question is:

Given a fixed budget of resettable-simulator CF branches, can a learned all-action landscape improve PPO learning at unqueried states, and can uncertainty allocate that CF budget better than uniform sampling?

The claimed contribution is *not* that advantage modification changes a policy. The proposed contribution is the combination of:

1. sparse all-action one-step CF supervision;
2. a policy-relative landscape student that affects unqueried rollout states; and
3. uncertainty- and leverage-aware allocation of later CF branches.

If all decision gates pass, the paper may claim an improved learning-counterfactual-cost frontier on the stated MiniGrid suite. It must not claim an exact oracle, universal PPO convergence, safe RL, or broad cross-domain generality. All tasks share MiniGrid’s partial symbolic observation and seven-action interface.

2 Definitions and notation

All CF targets, student outputs, uncertainty values, query decisions, and gates are detached from PPO’s actor gradient. The student is updated by supervised learning. The critic always uses the unperturbed PPO return target.

3 Theory: advantage perturbations steer only through actions

3.1 Shift relative to ordinary GAE PPO

Consider PPO’s unclipped one-step likelihood-ratio actor term:

$$g_{\text{GAE}}(\theta) = \mathbb{E} \left[\hat{A}_{\text{GAE}}(s, a) z_{\theta}(s, a) \right]. \quad (1)$$

For a detached perturbation $f(s, a)$, use

$$\tilde{A}(s, a) = \hat{A}_{\text{GAE}}(s, a) + f(s, a), \quad g_{\text{used}}(\theta) = \mathbb{E}[\tilde{A}(s, a) z_{\theta}(s, a)]. \quad (2)$$

Subtracting [equation \(1\)](#) from [equation \(2\)](#) gives

$$\boxed{g_{\text{used}}(\theta) - g_{\text{GAE}}(\theta) = \mathbb{E}[f(s, a) z_{\theta}(s, a)]}. \quad (3)$$

This is exact as a difference from the ordinary GAE score-gradient term; it does not require GAE itself to be unbiased for the true policy gradient.

Term or symbol	Definition
$s \in \mathcal{S}, o \in \mathcal{O}$	Full cloneable simulator state and policy observation. PPO observes $o = o(s)$, not necessarily all of s .
$a \in \mathcal{A} = \{1, \dots, K\}$	Discrete action; $K = 7$ in all selected MiniGrid tasks.
$\pi_\theta(a o), \pi(o)$	Categorical PPO policy and its K -vector of action probabilities. θ denotes actor parameters.
e_a	K -vector containing one at coordinate a and zero elsewhere.
$z_\theta(s, a)$	Score feature $\nabla_\theta \log \pi_\theta(a o(s))$.
$V_\phi(o)$	PPO critic, with parameters ϕ .
$\hat{A}_{\text{GAE}}(s, a)$	Ordinary sampled-action generalized advantage estimate.
γ	Discount factor used by PPO and the CF teacher.
$d(s, a)$	Terminal indicator after forced action a from s : one if terminal and zero otherwise.
$Q_{\text{CF}}(s, a)$	One-step CF critic-bootstrap value of forcing a at saved state s .
$y(s) = A_{\text{CF}}(s, \cdot)$	Centred K -vector teacher landscape made from all $Q_{\text{CF}}(s, a)$.
$B_{\psi_m}(o)$	Output of student head m , a K -vector predicted from observation.
$\bar{B}(s), \Sigma_B(s)$	Mean and covariance of the centred student-ensemble predictions.
$C_\pi(s)$	Categorical action covariance $\text{diag}(\pi(s)) - \pi(s)\pi(s)^\top$.
$A_{\text{dist}}(s, a)$	Centred scalar payload from the student mean for sampled action a .
$f_\psi(s, a)$	Detached advantage perturbation added only to the actor signal.
β	Global steering coefficient selected on development runs and then frozen.
$w(s)$	Confidence gate, in $[0, 1]$, reducing steering where the student is uncertain.
$u_{\text{ens}}(s)$	Policy-relevant epistemic uncertainty: ensemble prediction variance.
$N_{\text{abs}}(s)$	Visit count for an abstract-state/representation bucket; a coverage heuristic, not model uncertainty.
$\ell(s), q(s)$	Predicted policy leverage and final CF query priority.

Table 1: Definitions of every central quantity.

3.2 State-only perturbations cancel

For each state define the policy average and action-selective residual

$$\bar{f}(s) = \sum_{a=1}^K \pi(a | s) f(s, a), \quad \Delta f(s, a) = f(s, a) - \bar{f}(s). \quad (4)$$

Proposition 1 (Action-selectivity identity). *For a differentiable categorical policy,*

$$\mathbb{E}[f(s, a) z_\theta(s, a)] = \mathbb{E}[\Delta f(s, a) z_\theta(s, a)]. \quad (5)$$

Hence a state-only term $f(s, a) = c(s)$ has zero direct population score-gradient effect.

Proof. Conditioning on s ,

$$\mathbb{E}_{a \sim \pi}[z_\theta(s, a) | s] = \sum_a \pi(a | s) \nabla_\theta \log \pi(a | s) \quad (6)$$

$$= \nabla_\theta \sum_a \pi(a | s) = 0. \quad (7)$$

Therefore $\mathbb{E}[\bar{f}(s) z_\theta(s, a)] = 0$. Substitute $f = \bar{f} + \Delta f$ into [equation \(3\)](#). For $f = c(s)$, $\Delta f = 0$. $\square$

Remark 1. *The result is a population statement for the unclipped score term. PPO clipping, finite batches, repeated epochs, and batch-wise advantage normalization can create implementation-level effects. Actor advantage normalization is therefore disabled in every arm. E0 logs KL, clip fraction, and the realized gradient change.*

3.3 Categorical action-landscape geometry

Define the categorical score coordinate and its covariance:

$$\phi(s, a) = e_a - \pi(s), \quad C_\pi(s) = \text{diag}(\pi(s)) - \pi(s)\pi(s)^\top. \quad (8)$$

For any K -vector $b(s)$ of action preferences, the sampled-action scalar payload is

$$g_b(s, a) = b(s)^\top \phi(s, a) = b(s)_a - \pi(s)^\top b(s). \quad (9)$$

It is action-selective and conditionally centred:

$$\mathbb{E}[g_b(s, a) \mid s] = 0. \quad (10)$$

The induced logit-coordinate update is

$$\mathbb{E}[g_b(s, a)\phi(s, a) \mid s] = \mathbb{E}[\phi(s, a)\phi(s, a)^\top \mid s]b(s) \quad (11)$$

$$= C_\pi(s)b(s). \quad (12)$$

Thus $b(s)$ does not hard-code a chosen action. It specifies relative action preference, and the policy moves in the covariance-weighted direction $C_\pi b$. A constant vector is in the null space of C_π and has no direct actor effect.

4 Counterfactual teacher and landscape student

4.1 Teacher target

For a saved rollout state s , restore the full simulator state and force each action a . Let s'_a and $r(s, a)$ be the resulting successor and reward. The current CF-PPO implementation’s teacher is

$$Q_{\text{CF}}(s, a) = r(s, a) + \gamma[1 - d(s, a)]V_\phi(o(s'_a)). \quad (13)$$

It is a one-step counterfactual critic-bootstrap estimate. It is *not* an exact counterfactual return: its accuracy depends on critic approximation.

The all-action teacher landscape is centred under the old rollout policy:

$$y(s)_a = A_{\text{CF}}(s, a) = Q_{\text{CF}}(s, a) - \sum_{b=1}^K \pi_{\text{old}}(b \mid o(s))Q_{\text{CF}}(s, b). \quad (14)$$

The teacher is expensive because it requires restored-simulator branches for all actions at a labelled state.

4.2 Student target, prediction, and use

The student is a cheap observation-to-action-landscape function:

$$B_{\psi_m} : \mathcal{O} \rightarrow \mathbb{R}^K, \quad m = 1, \dots, M. \quad (15)$$

For the final method use $M = 5$ bootstrap ensemble heads. Heads have independent initialization and independent bootstrap masks over a rolling buffer $\mathcal{D}_{\text{CF}} = \{(o_i, y_i, \pi_i)\}$. The rolling buffer retains only recent labels because both critic and policy-relative targets drift.

For head m , define the masked supervised loss

$$\mathcal{L}_m(\psi_m) = \frac{1}{|\mathcal{D}_m|} \sum_{(o_i, y_i, \pi_i) \in \mathcal{D}_m} (B_{\psi_m}(o_i) - y_i)^\top C_{\pi_i}(s_i) (B_{\psi_m}(o_i) - y_i). \quad (16)$$

The C_π weighting ignores action-independent offsets and emphasizes errors that can influence a categorical PPO update.

At a current rollout state, centre each prediction under the *current* old policy:

$$B_m^c(s) = B_{\psi_m}(o(s)) - \mathbf{1} \pi_{\text{old}}(\cdot \mid o(s))^\top B_{\psi_m}(o(s)). \quad (17)$$

Define

$$\bar{B}(s) = \frac{1}{M} \sum_{m=1}^M B_m^c(s), \quad A_{\text{dist}}(s, a) = \bar{B}(s)_a. \quad (18)$$

Equation [equation \(17\)](#) ensures that $A_{\text{dist}}(s, a) = B_{\text{raw}}(s)_a - \pi^\top B_{\text{raw}}(s)$, so even a state-conditioned network produces an action-selective payload.

4.3 Why student recovery matters

At state s , direct CF induces $C_\pi(s)y(s)$, while the student induces $C_\pi(s)\bar{B}(s)$. Their difference is exactly

$$\boxed{\delta g_{\text{student}}(s) - \delta g_{\text{teacher}}(s) = C_\pi(s)(\bar{B}(s) - y(s))}. \quad (19)$$

The held-out policy-update error and directional alignment are

$$\epsilon_{\text{PG}}(s) = \|C_\pi(s)(\bar{B}(s) - y(s))\|_2, \quad \langle C_\pi(s)\bar{B}(s), C_\pi(s)y(s) \rangle. \quad (20)$$

These metrics test whether the student has recovered a useful action direction on states never used as CF labels.

5 PPO integration, ensemble variance, and CF selection

The proposed actor-only perturbation is

$$f_\psi(s, a) = \beta w(s) A_{\text{dist}}(s, a), \quad \tilde{A} = \hat{A}_{\text{GAE}} + f_\psi. \quad (21)$$

$\beta \geq 0$ is selected on separate development runs and then frozen. We use an additive form, rather than a raw interpolation that also weakens GAE.

For PPO ratio

$$r_\theta(s, a) = \frac{\pi_\theta(a \mid o(s))}{\pi_{\text{old}}(a \mid o(s))}, \quad (22)$$

the actor loss is

$$\mathcal{L}_{\text{actor}} = -\mathbb{E} \left[\min \left(r_\theta \tilde{A}, \text{clip}(r_\theta, 1 - \epsilon, 1 + \epsilon) \tilde{A} \right) \right]. \quad (23)$$

The critic loss uses ordinary unperturbed GAE returns only.

The ensemble covariance of current centred predictions is

$$\Sigma_B(s) = \frac{1}{M-1} \sum_{m=1}^M (B_m^c(s) - \bar{B}(s))(B_m^c(s) - \bar{B}(s))^\top. \quad (24)$$

The ensemble-variance uncertainty is

$$\boxed{u_{\text{ens}}(s) = \sqrt{\text{tr}[C_{\pi}(s)\Sigma_B(s)]}.} \quad (25)$$

It is the prediction variance in action coordinates PPO can use, not generic raw output variance. The confidence gate is

$$w(s) = \exp[-u(s)/\tau], \quad u(s) = u_{\text{ens}}(s) + c_{\text{count}}u_{\text{count}}(s), \quad (26)$$

where $u_{\text{count}}(s) = 1/\sqrt{N_{\text{abs}}(s) + 1}$ is a separate novelty or coverage heuristic, not ensemble uncertainty.

Predicted policy leverage is

$$\ell(s) = \sqrt{\bar{B}(s)^\top C_{\pi}(s) \bar{B}(s) + \varepsilon_{\text{num}}}, \quad (27)$$

where $\varepsilon_{\text{num}} > 0$ is a numerical stabilizer. Select CF states using

$$\boxed{q(s) = [u_{\text{ens}}(s) + c_{\text{count}}u_{\text{count}}(s)]\ell(s).} \quad (28)$$

After ranking, impose diversity in encoder-feature space to avoid spending all branches on visually similar navigation states.

6 CF budget and valid controls

Set $p_{\text{label}} = 0.02$: label two percent of rollout states with all seven actions. The code must report actual CF branch transitions. A rough upper estimate is $7p_{\text{label}} = 14\%$ extra one-step simulator work, but the reported number from the implementation is authoritative.

For a per-state random action permutation τ_s , use the properly re-centred control

$$y_{\text{perm}}(s, a) = y(s, \tau_s(a)) - \sum_b \pi_{\text{old}}(b \mid s) y(s, \tau_s(b)). \quad (29)$$

For the wrong-state student, permute teacher vectors across states with permutation σ , then re-centre under the destination policy:

$$y_{\text{wrong}}(s_i) = y(s_{\sigma(i)}) - \mathbf{1} \pi_{\text{old}}(\cdot \mid s_i)^\top y(s_{\sigma(i)}). \quad (30)$$

Re-centring is essential: otherwise a control contains an unintended state-only offset.

7 Environment suite

Unlock terminates on opening the locked door; RedBlueDoors requires red before blue and terminates on a blue-first opening; UnlockPickup ends on correct-box pickup [3, 4, 5]. These task differences make the suite more informative than a single DoorKey size, while remaining an honest within-MiniGrid generalization test.

8 Minimal experiment plan

Every final comparison uses five paired random seeds. Within a seed, arms share policy initialization and environment seed schedule. β , τ , the student buffer horizon, c_{count} , and diversity threshold are set using separate DoorKey-5x5 development runs and frozen before the five final seeds.

Environment	Prerequisite structure	Role
MiniGrid-DoorKey-5x5-v0	Key $\rightarrow$ locked door $\rightarrow$ goal	Fast development and E0 mechanism check.
MiniGrid-DoorKey-6x6-v0	Same structure, longer navigation	In-family scale validation.
MiniGrid-Unlock-v0	Key $\rightarrow$ open locked door	Simple external teacher sanity check.
MiniGrid-RedBlueDoors-6x6-v0	Open red, then blue; blue first fails	Sequential ordering and irreversible decision task.
MiniGrid-UnlockPickup-v0	Unlock door $\rightarrow$ enter room $\rightarrow$ pick up box	Compositional prerequisite task.

Table 2: Environment ladder. DoorKey-8x8 is excluded until clean PPO solves it reliably.

Stage	Environments	Seeds	Conditions and required conclusion
E0	DoorKey-5x5	5 paired	GAE, matched state-only perturbation, action-selective direct CF. Establish the selectivity mechanism and audit PPO wiring.
E1	DoorKey-6x6; Unlock; RedBlueDoors-6x6; Unlock-Pickup	5 each	GAE, full direct CF, action-permuted CF. Establish that correct current CF carries useful action semantics across task structures.
E2	DoorKey-6x6; RedBlueDoors-6x6; UnlockPickup	5 each	GAE, sparse direct CF, uniform distilled CF, wrong-state student. Establish amortization at equal CF branch budget.
E3	RedBlueDoors-6x6; Unlock-Pickup	5 each	Uniform distillation, uncertainty-only querying, uncertainty-times-leverage querying. Establish value of active allocation at equal budget.

Table 3: Four-stage experimental sequence.

8.1 E0: mechanism

From the same stochastic checkpoint, compare

$$\hat{A}_{\text{GAE}}, \quad \hat{A}_{\text{GAE}} + \beta c(s), \quad \hat{A}_{\text{GAE}} + \beta y(s)_a. \quad (31)$$

$c(s)$ is identical for all actions and RMS-matched to $y(s)_a$ on calibration rollouts. E0 is a one-to-few-update diagnostic, not a headline learning run. Log empirical perturbation gradient, KL, entropy, clip fraction, and actor loss. The state-only effect should be near the GAE null; direct CF should have a nonzero action-aligned effect.

8.2 E1: teacher viability

Full direct CF is an expensive ceiling, not the final method. The required early-learning ordering is

$$\text{direct CF} > \text{GAE} \quad \text{and} \quad \text{direct CF} > \text{action-permuted CF} \quad (32)$$

in a majority of the four tasks. If the permutation is equally good, the signal may only be a generic action prior or entropy effect and should not be described as state-local CF credit.

Condition	At queried states	At unqueried states
GAE PPO	GAE	GAE
Sparse direct CF	Direct all-action CF actor contribution	GAE
Uniform distilled CF	Add (o, y) to the student buffer	Student-guided signal from equation (21)
Wrong-state student	Add (o, y_{wrong}) to the buffer	Mismatched student-guided signal

Table 4: Equal-budget E2 conditions.

8.3 E2: sparse-label amortization

All CF conditions get exactly the same $p_{\text{label}} = 2\%$ label fraction.

The primary test is

$$\boxed{\text{uniform distilled CF} > \text{sparse direct CF}} \quad \text{at equal CF branch cost.} \quad (33)$$

This shows that the student spreads sparse CF information beyond the queried states. Separately report [equation \(20\)](#) on held-out CF-labelled states that never enter the training buffer, including a split between queried and never-queried evaluation states.

8.4 E3: active CF allocation

At the same label and branch budget as E2, compare uniform state selection, largest u_{ens} selection, and the full score in [equation \(28\)](#). The primary test is

$$\boxed{\text{uncertainty-times-leverage} > \text{uniform}} \quad \text{at equal CF branch cost.} \quad (34)$$

After querying, selected states should have higher realised student error or teacher correction magnitude than uniform states. Otherwise a performance difference cannot be credited to informative allocation.

9 Metrics, plots, and decision gates

Category	Required measurement
Primary learning	Success AUC, final success, and transitions to 50% and 90% success.
Task progress	DoorKey: key pickup and door opening. Unlock: door opening. RedBlueDoors: red opening, valid blue-after-red success, invalid blue-first termination. UnlockPickup: key pickup, door opening, correct-box pickup.
PPO dynamics	Entropy, approximate KL, clip fraction, actor loss, critic loss, and explained variance if available.
Landscape recovery	Held-out ϵ_{PG} , action-ranking agreement, and alignment in equation (20) .
Budget honesty	On-policy transitions, labelled states, actual CF branch transitions, total simulator transitions, and wall-clock time.

Table 5: Required readouts.

Plot every learning curve against both on-policy transitions and total simulator transitions including CF branches. Show five faint seed traces, mean, and a nonparametric 95% bootstrap

confidence band. For final scalar comparisons, report paired seed-level differences and confidence intervals. Any CF labels used solely as a held-out audit are reported separately and are never added to the student buffer.

Result	Interpretation and action
Clean PPO fails to solve an environment	Exclude it until the baseline is fixed; it is not evidence about CF.
State-only perturbation strongly changes learning	Audit clipping, normalization, repeated epochs, and critic wiring before making the theorem claim.
Direct CF fails to beat GAE or permuted CF	The current one-step teacher is not useful enough to distil; stop the pipeline.
Distilled CF fails to beat sparse direct CF	No evidence of useful amortization.
Wrong-state student matches correct student	Reframe benefit as generic action-prior/exploration shaping; do not claim state-local landscape recovery.
Active selection fails to beat uniform	Retain uniform distillation only; active allocation is unsupported.
Gain is explained solely by faster entropy collapse	Add an entropy-matched GAE control before attributing gain to CF credit.

Table 6: Predeclared interpretation gates.

10 Supported conclusion if all gates pass

The justified conclusion would be:

Across evaluated MiniGrid tasks with key-door, ordered-door, and unlock-pickup prerequisite structures, a fixed budget of one-step all-action counterfactual branches can be distilled into a policy-relative action landscape that guides PPO at unqueried states. Ensemble variance and predicted policy leverage can improve the return obtained per counterfactual branch over uniform querying.

This is deliberately limited to a critic-bootstrapped teacher and the stated task family. It does not equate CF with exact long-horizon counterfactual return and does not establish universal PPO improvement.

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